

Notes: Card, Heining, and Kline (QJE, 2013)

Workplace Heterogeneity and the Rise of West German Wage Inequality

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1 Big picture

Card, Heining, and Kline study why wage inequality rose so much in West Germany between the mid-1980s and the late 2000s. Their key point is that wage inequality is not only a worker-skill story. It is also a workplace story: some establishments pay systematically higher wage premiums than others, and workers are increasingly sorted so that high-wage workers work at high-wage establishments.

The paper fits Abowd-Kramarz-Margolis style two-way fixed effect models separately in four overlapping intervals: 1985–1991, 1990–1996, 1996–2002, and 2002–2009. The estimated wage for worker i in year t is decomposed into a worker fixed effect, an establishment fixed effect, observable time-varying controls, and a residual. The main finding is that the dramatic rise in West German wage inequality is explained by three forces: rising dispersion in person effects, rising dispersion in establishment wage premiums, and rising covariance between the two.

The most important conceptual lesson is that establishment wage premiums are not a small nuisance component. They become a quantitatively central part of inequality. The paper shows that the variance of establishment effects rises, and even more importantly, the correlation between person and establishment effects rises sharply. In other words, the labor market becomes more assortative: workers who would earn high wages anywhere are increasingly employed at firms that pay high wage premiums to all workers.

2 Incremental contribution of the paper

1. **Moves the inequality debate beyond worker-only explanations.** Earlier work emphasized supply and demand for skills, technology, trade, and institutions. This paper shows that establishment wage premiums and worker-firm sorting are first-order components of the inequality increase.
2. **Uses administrative matched employer-employee data at massive scale.** The IEB social-security data allow the authors to follow millions of workers and establishments over time and to estimate worker and establishment fixed effects in very large connected sets.
3. **Tests whether the AKM additive structure is a good approximation.** The paper does not merely assume that wages are additively separable in worker and establishment effects. It uses event studies and residual diagnostics to check whether job-match effects or endogenous mobility are likely to invalidate the model. The additive model performs surprisingly well.
4. **Shows that assortative matching matters.** The covariance between worker and establishment effects rises from a negligible share of wage variance to a large component of inequality. This is a key contribution because inequality grows not only because workers and firms become more heterogeneous separately, but because they match differently.
5. **Decomposes between-group wage inequality using worker and establishment components.** The paper shows that changes in mean establishment effects explain large shares of the widening education, occupation, and industry wage gaps.
6. **Connects rising establishment heterogeneity to institutional change.** New establishments after the mid-1990s display more dispersion in wage premiums, and fewer new establishments are covered by collective bargaining agreements. This suggests that wage-setting

institutions may be related to the rise in firm-level heterogeneity.

3 Research question

The central question is: *How much of the rise in West German wage inequality is due to worker heterogeneity, establishment heterogeneity, and the sorting of workers across establishments?*

This question is important because standard wage inequality decompositions often attribute wage growth to individual characteristics or residual inequality. But if firms differ persistently in the wage premiums they pay, and if workers are nonrandomly allocated across firms, then the wage distribution reflects both worker skills and workplace pay policies. A high-wage worker at a high-wage firm earns more than the sum of either advantage alone in the aggregate variance decomposition because worker and firm components become positively correlated.

4 Data and institutional setting

The paper uses the Integrated Employment Biographies data from the German social security system. These data contain annual information on earnings, days worked, establishment identifiers, occupation, industry, education, and full-time/part-time status. The preferred sample is full-time male workers ages 20–60 in West Germany, excluding marginal jobs and training jobs. The authors also analyze full-time women as a robustness check.

The data are well suited to the question because they link workers to employers over time. This allows the authors to identify person effects from repeated observations of the same worker and establishment effects from workers who move between establishments. The estimation is performed within the largest connected set of establishments and workers in each interval. This connected-set requirement is crucial because fixed effects are identified only relative to other fixed effects connected by worker mobility.

Germany is a particularly useful setting because its wage inequality rose sharply after the mid-1990s, and the rise coincided with labor-market reforms and a decline in collective bargaining coverage. The paper does not claim that reforms alone caused the rise, but it shows that establishment-level wage heterogeneity is closely related to the timing and institutional context of the German labor market.

5 Econometric model

In each interval, the paper estimates the additive AKM wage equation:

$$y_{it} = \alpha_i + \psi_{J(i,t)} + x'_{it}\beta + r_{it},$$

where y_{it} is log real daily wage, α_i is a person effect, $\psi_{J(i,t)}$ is the establishment effect for the establishment employing worker i in year t , $x'_{it}\beta$ includes observable controls such as year effects interacted with education and age profiles, and r_{it} is the residual.

The interpretation is as follows. The person effect captures portable earnings capacity: skills or worker attributes that are rewarded similarly across employers. The establishment effect captures

a workplace wage premium: the part of pay that workers receive because they work at a particular establishment. The residual captures transitory or match-specific variation not explained by the additive model.

A central variance decomposition is:

$$\text{Var}(y) = \text{Var}(\alpha) + \text{Var}(\psi) + 2 \text{Cov}(\alpha, \psi) + \text{Var}(x\beta) + \text{Var}(r) + 2 \text{Cov}(x\beta, \alpha + \psi).$$

The paper’s main empirical result is that the first three terms - person variance, establishment variance, and their covariance - account for nearly all of the increase in male wage inequality.

6 Identification logic and model validation

The AKM model requires that worker mobility provides useful comparisons across establishments. If movers are systematically selected based on match-specific shocks, the establishment effects could be biased. The authors therefore use event-study evidence around job moves. Workers who move from low-wage to high-wage establishments experience wage gains, while workers who move in the opposite direction experience approximately symmetric losses. Wages are also relatively flat before moves, showing little evidence of Ashenfelter-style pre-move dips. These patterns support the additive fixed-effect interpretation.

The paper also compares the AKM model to a more flexible match model with a separate fixed effect for each worker-establishment pair. The match model improves fit only modestly. This indicates that match-specific components exist, but they are much smaller and more stable than the person and establishment effects that drive inequality.

7 Main findings

7.1 Wage inequality rises sharply after the mid-1990s

The raw wage distribution shows strong divergence. Upper percentiles rise or remain stable, while lower percentiles decline strongly after the mid-1990s. The gap between the 20th and 80th percentiles expands by about 20 log points between 1996 and 2009.

7.2 Residual inequality remains important after controls

Demographic and occupation controls reduce the level of measured dispersion, but they do not remove the strong upward trend. This means that rising inequality is not simply a composition effect from aging, education, occupation, or industry changes.

7.3 Establishment wage premiums become more dispersed

The standard deviation of establishment effects increases from 0.159 in 1985–1991 to 0.230 in 2002–2009. This is a large increase in the workplace component of wages. The increase is especially important because establishment effects are estimated net of person effects.

7.4 Worker-firm sorting becomes much more positive

The correlation between person and establishment effects rises from 0.034 in the first interval to 0.249 in the last interval. This is one of the paper’s most important results. It means that high person-effect workers are increasingly concentrated in high establishment-effect workplaces.

7.5 The match component is small and stable

The match model residual RMSE is smaller than the AKM residual RMSE, but the implied standard deviation of the match effect is modest, rising from about 0.060 to 0.075. This suggests that idiosyncratic worker-firm match quality is not the main reason wage inequality rises.

7.6 New establishments and declining collective bargaining matter

The rise in establishment heterogeneity is concentrated among newer establishments. Establishments born after the mid-1990s have more dispersed wage premiums, and they are less likely to be covered by collective agreements. This is consistent with the view that institutional wage compression weakened, especially for new firms.

8 Figures and tables

8.1 Figure I: Trends in Percentiles of Real Log Daily Wages for West German Men

Read. Figure I plots the 10th, 20th, 50th, and 80th percentiles of real log daily wages for full-time West German men, normalized to their 1996 values. The lower percentiles fall sharply after the mid-1990s, especially the 10th percentile, while the 80th percentile rises. The median is comparatively flat.

Interpretation. This figure motivates the entire paper. The increase in inequality is not just top-end growth; it is also substantial deterioration at the bottom. The timing matters because the acceleration occurs around the period when German labor-market institutions were changing. The pattern calls for an explanation that can account for both weak lower-tail wage growth and increasing upper-tail advantage.

8.2 Table I: Summary Statistics for Samples of Full-Time Men and Women

Read. Table I reports sample sizes and wage moments for full-time men and women every five years. The male allocated wage standard deviation rises from 0.429 in 1985 to 0.586 in 2009. The female allocated wage standard deviation rises from 0.470 to 0.606. Censoring is much more important for men than women, which is why the paper imputes top-coded wages.

Interpretation. The table establishes that the inequality increase is visible in basic moments before any complicated model is estimated. It also clarifies the measurement challenge: administrative wages are top-coded at the social security maximum, so the authors use a stochastic Tobit

allocation for censored observations. The broad similarity of rising dispersion for men and women suggests that the phenomenon is not limited to one gender sample.

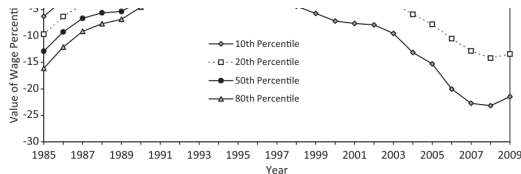


FIGURE I
Trends in Percentiles of Real Log Daily Wages for West German Men
Figure shows percentiles of log real daily wage for full-time male workers on their main job, deviated from value of same percentile in 1996 and multiplied by 100.

Figure I: Trends in Percentiles of Real Log Daily Wages for West German Men

TABLE I
SUMMARY STATISTICS FOR SAMPLES OF FULL-TIME MEN AND WOMEN

	(1) Number observations	Log real wage, unallocated		(4) Percent censored	Log real wage, allocated	
		(2)	(3)		(5)	(6)
		Mean	Std. dev.		Mean	Std. dev.
<i>Panel A. Full-time men</i>						
1985	11,980,159	4.221	0.387	10.63	4.247	0.429
1990	13,289,988	4.312	0.398	11.92	4.342	0.445
1995	13,101,809	4.340	0.415	9.78	4.361	0.447
2000	12,930,046	4.327	0.464	10.31	4.352	0.502
2005	11,857,526	4.310	0.519	9.36	4.336	0.562
2009	12,104,223	4.277	0.535	10.00	4.308	0.586
<i>Panel B. Full-time women</i>						
1985	6,068,863	3.836	0.462	1.52	3.840	0.470
1990	7,051,617	3.942	0.476	2.01	3.947	0.486
1995	7,030,596	4.026	0.483	1.95	4.030	0.491
2000	7,009,075	4.019	0.532	2.47	4.026	0.545
2005	6,343,006	3.999	0.573	2.36	4.006	0.588
2009	6,566,429	3.979	0.587	2.80	3.988	0.606

Notes. Samples includes employees in West Germany age 20–60, working full-time in nonmarginal jobs. Real wage is based on average daily earnings at the full-time job with highest total earnings during the calendar year, adjusted for inflation using the Consumer Price Index. Unallocated wage data in columns (2) and (3) are based on raw daily wage data, which are censored at social security maximum for the corresponding year. Allocated wage data in columns (5) and (6) include stochastic allocation of censored observations based on a Tobit model.

Table I: Summary Statistics for Samples of Full-Time Men and Women

8.3 Table II: Summary Statistics for Overall Sample and Individuals in Largest Connected Set

Read. Table II describes the four estimation intervals and compares the full sample to the largest connected set. The largest connected set covers about 97–98% of person-year observations and about 96% of individuals in each interval. The standard deviation of log wages in the connected set rises from 0.370 to 0.499.

Interpretation. This table is important for credibility. Two-way fixed effects are identified only within connected components. If the connected set were small or selected, the decomposition would be less persuasive. Instead, the connected set contains almost the whole sample, so the estimation sample is economically representative of the full-time male labor market. The increase in the connected-set wage standard deviation is nearly identical to the full-sample increase.

8.4 Figure II: Trends in Wage Inequality for Full-Time Male Workers

Read. Figure II plots the standard deviation of log wages and normalized percentile gaps. The 80–20, 80–50, and 50–20 gaps all rise. Inequality growth is especially steep after the mid-1990s.

Interpretation. This figure shows that the rise in inequality is not an artifact of a single statistic or of top-coding corrections. Percentile gaps are robust distributional measures, and they move similarly to the standard deviation. The rise is also not confined to either the upper or lower tail: both the upper-tail and lower-tail gaps widen, which implies a broad fanning out of wages.

Interval	All full-time men, age 20-60				Individuals in largest connected set			
	(1) Number person/yr. obs.	(2) Number individuals	Log real daily wage		(5) Number person/yr. obs.	(6) Number individuals	Log real daily wage	
			(3) Mean	(4) Std. dev.			(7) Mean	(8) Std. dev.
1985-1991 Ratio: largest connected/all	86,230,997	17,021,779	4.344	0.379	84,185,730 97.6	16,295,106 95.7	4.351 100.2	0.370 97.7
1990-1996 Ratio: largest connected/all	90,742,309	17,885,361	4.391	0.392	88,662,398 97.7	17,223,290 96.3	4.398 100.2	0.384 97.9
1996-2002 Ratio: largest connected/all	85,853,626	17,094,254	4.397	0.439	83,699,582 97.5	16,384,815 95.8	4.405 100.2	0.432 98.3
2002-2009 Ratio: largest connected/all Change from first to last interval	93,037,963	16,553,835	4.387	0.505	90,615,841 97.4	15,834,602 95.7	4.397 100.2	0.499 98.8 0.045 0.128

Notes: Sample consists of full-time male workers ages 20-60 employed in nonmarginal jobs and not currently in training. Daily wage is imputed for censored observations using

Table II: Summary Statistics for Overall Sample and Individuals in Largest Connected Set

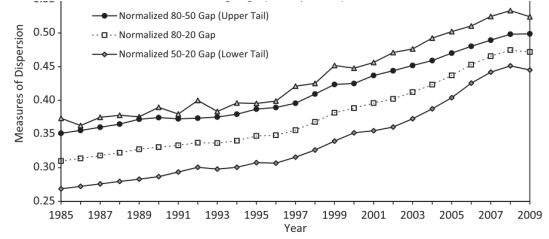


FIGURE II
Trends in Wage Inequality for Full-Time Male Workers

This figure shows measures of dispersion in real daily wage for full-time male workers. Normalized percentile gaps are differences in percentiles divided by corresponding differences in percentiles of standard normal variate.

Figure II: Trends in Wage Inequality for Full-Time Male Workers

8.5 Figure III: Wage Inequality Trends for Alternative Samples of Workers

Read. Figure III compares normalized 80–20 wage gaps for all male workers, full-time male workers, all female workers, and full-time female workers. All four series trend upward, with full-time women displaying high dispersion and an upward pattern similar to men.

Interpretation. The figure shows that the rise in inequality is not an artifact of restricting the sample to full-time men. The authors focus on full-time men because hours are not observed in the administrative data and full-time work avoids part-time hours confounds. But the similar patterns for other samples strengthen the argument that West German wage inequality rose broadly.

8.6 Figure IV: Raw and Residual Standard Deviations from Alternative Wage Models

Read. Figure IV shows raw wage dispersion and residual dispersion after successively adding age/education controls, occupation controls, and industry controls. Each control set lowers the level of dispersion, but the upward trend remains strong.

Interpretation. This figure rules out a simple observable-composition story. Age, education, occupation, and industry explain part of wage-level differences, but they cannot explain the timing or magnitude of the rise in dispersion. The remaining trend is exactly where the worker-establishment decomposition becomes valuable: the unexplained component may reflect person heterogeneity, establishment premiums, and sorting.



FIGURE III
Wage Inequality Trends for Alternative Samples of Workers

Based on tabulations of SIAB. Measured wage is average daily wage in job with highest total earnings in the year. Wage gap is the difference between the 80th percentile of log real wages and the 20th percentile, divided by 80–20 gap

Figure III: Wage Inequality Trends for Alternative Samples of Workers

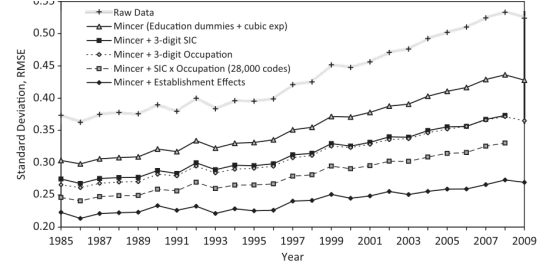


FIGURE IV
Raw and Residual Standard Deviations from Alternative Wage Models

See note to Figure II. Figure shows measures of dispersion in actual and residual real daily wage for full-time male workers. Residual wage is residual from linear regression model. "Mincer" refers to model with dummies for education categories and cubic in experience, fit separately in each year. Other models add controls as indicated.

Figure IV: Raw and Residual Standard Deviations from Alternative Wage Models

8.7 Figure V: Mean Wages of Job Changers Classified by Quartile of Mean Wage of Coworkers at Origin and Destination Establishments for 1985–1991 and 2002–2009

Read. Figure V shows event-study wage profiles around job moves. Workers moving from low-coworker-wage establishments to high-coworker-wage establishments gain wages, while workers moving in the opposite direction lose wages. Profiles are relatively flat before the move.

Interpretation. This figure is a key validation of the AKM model. Symmetric gains and losses imply that establishment wage premiums are real: moving to a higher-wage establishment raises pay, and moving away lowers pay. The absence of strong pre-move wage trends reduces concern that wage changes are driven by transitory shocks that cause mobility. The figure also suggests that match-specific components are not dominant, because workers moving in opposite directions experience roughly opposite wage changes.

8.8 Table III: Estimation Results for AKM Model, Fit by Interval

Read. Table III reports the estimated AKM components. The standard deviation of person effects rises from 0.289 to 0.357. The standard deviation of establishment effects rises from 0.159 to 0.230. The correlation between person and establishment effects rises from 0.034 to 0.249. The adjusted R^2 of the AKM model is high, rising from 0.896 to 0.927.

Interpretation. This table contains the core evidence. Both worker heterogeneity and workplace heterogeneity rise, but the sorting result is especially striking: the correlation between person and establishment effects becomes much more positive. The match model fits better than AKM, but only modestly, and the implied match component is small relative to the main components. Therefore the additive decomposition is informative, not merely a statistical convenience.

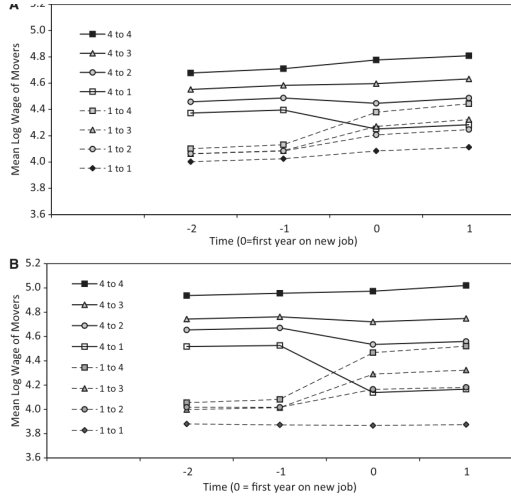


FIGURE V

Mean Wages of Job Changers Classified by Quartile of Mean Wage of Coworkers at Origin and Destination Establishment (A) 1985–1991, (B) 2002–2009

Figure shows mean wages of male workers observed in 1985–1991 or 2002–2009 who change jobs in the respective interval, and held the preceding job for two or more years, and the new job for two or more years. “Job” refers to establishment with most earnings in year, excluding part-time work. Each job is classified into quartiles based on mean wage of coworkers.

Figure V: Mean Wages of Job Changers Classified by Quartile of Mean Wage of Coworkers at Origin and Destination Establishments for 1985–1991 and 2002–2009

	(1) Interval 1 1985–1991	(2) Interval 2 1990–1996	(3) Interval 3 1996–2002	(4) Interval 4 2002–2009
Person and establishment parameters				
Number person effects	16,295,106	17,223,290	16,384,815	15,834,892
Number establishment effects	1,221,098	1,357,824	1,476,705	1,504,095
Summary of parameter estimates				
Std. dev. of person effects (across person-year obs.)	0.289	0.304	0.327	0.357
Std. dev. of establ. Effects (across person-year obs.)	0.159	0.172	0.194	0.230
Std. dev. of X_b (across person-year obs.)	0.121	0.088	0.093	0.084
Correlation of person/establ. Effects (across person-year obs.)	0.034	0.097	0.169	0.249
Correlation of person effects/ X_b (across person-year obs.)	-0.051	-0.102	-0.063	0.029
Correlation of establ. effects/ X_b (across person-year obs.)	0.057	0.039	0.050	0.112
RMSE of AKM residual	0.119	0.121	0.130	0.135
Adjusted R-squared	0.896	0.901	0.909	0.927
Comparison match model				
RMSE of match model	0.103	0.105	0.108	0.112
Adjusted R^2	0.922	0.925	0.937	0.949
Std. dev. of match effect*	0.060	0.060	0.072	0.075
Addendum				
Std. dev. log wages	0.370	0.384	0.432	0.499
Sample size	84,185,730	88,662,398	83,699,582	90,615,841

Note: Results from OLS estimation of equation (1). See notes to Table II for sample composition. X_b includes year dummies interacted with education dummies, and quadratic and cubic terms in age interacted with education dummies. Total of 29 parameters in intervals 1–3, 44 in interval 4. Match model includes X_b and separate dummy for each job (person-establishment pair).

*Standard deviation of match effect estimated as square root of difference in mean squared errors between AKM model and match effect model.

Table III: Estimation Results for AKM Model, Fit by Interval

8.9 Figure VI: Mean Residuals by Person/Establishment Deciles, 2002–2009

Read. Figure VI plots mean residuals from the AKM model by deciles of estimated person effects and establishment effects. The residuals are small and do not show large systematic failures of fit across the two-dimensional worker-firm distribution.

Interpretation. This is another model-diagnostic figure. A serious violation of additivity would show up as large residuals for particular worker-firm cells; for example, high-person-effect workers might earn especially large match premiums at high-premium firms. The small residuals imply that the additive model captures the main wage structure even in the cells where sorting matters most.

8.10 Figure VII: Mean Wages of Movers Classified by Quartile of Establishment Effects for Origin and Destination Firms, 2002–2009

Read. Figure VII repeats the mover event study using estimated establishment effects rather than coworker mean wages. The wage changes closely match predicted changes based on movement between establishment-effect quartiles.

Interpretation. This figure provides direct support for interpreting ψ_j as an establishment wage premium. Moving from a low- ψ to a high- ψ establishment increases wages; moving from high to low decreases wages. The parallel and symmetric profiles are hard to reconcile with a pure worker-selection story and are much more consistent with actual establishment-level pay premiums.

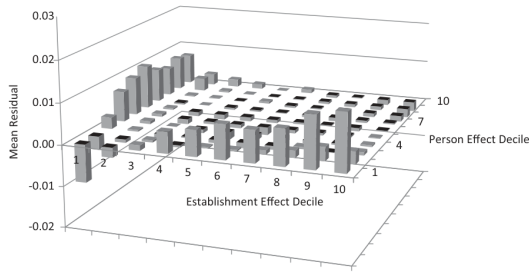


FIGURE VI
Mean Residuals by Person/Establishment Deciles, 2002–2009

Figure shows mean residuals from estimated AKM with cells defined by decile of estimated establishment effect, interacted with decile of estimated person effect. See column (4) of Table III for summary of model parameters.

Figure VI: Mean Residuals by Person/Establishment Deciles, 2002–2009

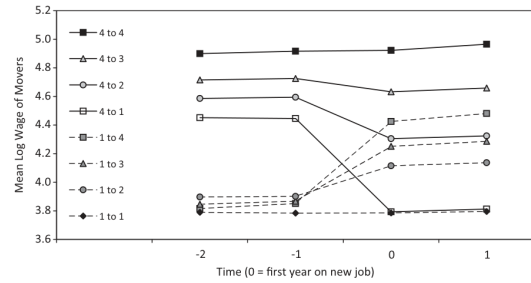


FIGURE VII
Mean Wages of Movers Classified by Quartile of Establishment Effects for Origin and Destination Firms, 2002–2009

Figure shows mean wages of male workers observed in 2002–2009 who change jobs in the interval and held the preceding job for two or more years, and the new job for two or more years. “Job” refers to main job in year, excluding part-time jobs. Each job is classified into quartiles based on estimated establishment effect from AKM model presented in Table III column 4.

Figure VII: Mean Wages of Movers Classified by Quartile of Establishment Effects for Origin and Destination Firms, 2002–2009

8.11 Figure VIII: Joint Distribution of Person and Establishment Effects, 1985–1991 and 2002–2009

Read. Figure VIII shows the joint distribution of person-effect deciles and establishment-effect deciles in the first and last intervals. The first interval is closer to flat, while the last interval shows more mass along the positive diagonal: high-person-effect workers are more often in high-establishment-effect workplaces.

Interpretation. This is the visual version of the rising covariance result. It shows that assortative matching is not just a statistic but a distributional change. The labor market increasingly pairs high-wage workers with high-wage firms and low-wage workers with low-wage firms. This magnifies inequality because worker and workplace advantages reinforce each other.

8.12 Table IV: Decomposition of the Rise in Wage Inequality

Read. Table IV decomposes the variance of male log wages between interval 1 and interval 4. Total variance rises by 0.112. The variance of person effects explains 39% of the rise, establishment effects explain 25%, and the covariance between person and establishment effects explains 34%. The variance of observables falls and therefore does not explain the rise.

Interpretation. Table IV is the paper’s headline quantitative decomposition. The rise in wage inequality is almost exactly a combination of worker heterogeneity, establishment heterogeneity, and sorting. The counterfactual rows show that if the person-establishment correlation had not risen, the increase in variance would be much smaller; if establishment-effect dispersion had also not risen, the increase would be less than half of the actual rise. Workplace forces are therefore not secondary.

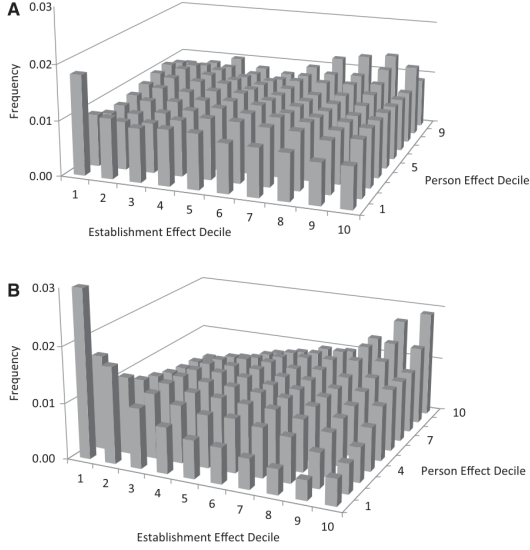


FIGURE VIII
Joint Distribution of Person and Establishment Effects (A) 1985–1991, (B) 2002–2009

Figure shows joint distribution of estimated person and establishment effects from AKM model for respective intervals. See Table III columns (1) and (4) for summary of model parameters for 1985–1991 and 2002–2009.

Figure VIII: Joint Distribution of Person and Establishment Effects, 1985–1991 and 2002–2009

TABLE IV
DECOMPOSITION OF THE RISE IN WAGE INEQUALITY

	Interval 1 (1985–1991)		Interval 4 (2002–2009)		Change from interval 1 to 4	
	(1) Var. component	(2) Share of total	(3) Var. component	(4) Share of total	(5) Var. component	(6) Share of total
Total variance of log wages	0.137	100.0	0.249	100.0	0.112	100
Components of variance:						
Variance of person effect	0.084	61.3	0.127	51.2	0.043	39
Variance of establ. effect	0.025	18.5	0.053	21.2	0.027	25
Variance of X_b	0.015	10.7	0.007	2.8	–0.008	–7
Variance of residual	0.011	8.2	0.015	5.9	0.003	3
$\text{Cov}(\text{person}, \text{establ.})$	0.003	2.3	0.041	16.4	0.038	34
$\text{Cov}(X_b, \text{person} + \text{establ.})$	–0.001	–1.0	0.006	2.4	0.007	7
Counterfactuals for variance of log wages*						
1. No rise in correl. of person/estab. effects	0.137		0.213		0.077	69
2. No rise in var. of establ. effect	0.137		0.209		0.072	64
3. Both 1 and 2	0.137		0.184		0.047	42

Notes: See notes to Table II for sample composition. Calculations based on estimated AKM models summarized in Table III. Entry in column (5) is change in variance component from interval 1 to interval 4. Entry in column (6) is ratio of the change in the variance component to the total change in variance of wages reported in first row of table (as a percentage).
*Counterfactual 1 computes the counterfactual rise in variance assuming the correlation between the person and establishment effects remains at its interval 1 value—that is, imposing the restriction that $\text{Cov}(\text{person}, \text{establ.}) = \rho_1 \sqrt{\text{Var}(\text{person}) \times \text{Var}(\text{establ.})}$ where the subscript 4 refers to the interval 4 value of the statistic and ρ_1 is the correlation between the person and establishment effects in interval 1. Counterfactual 2 assumes that the variance of establishment effects remains at its interval 1 level. Counterfactual 3 imposes both restrictions.

Table IV: Decomposition of the Rise in Wage Inequality

8.13 Table V: Decomposition of Changes in Relative Wages by Education Level, 1985–1991 versus 2002–2009

Read. Table V decomposes changes in mean wages by education group relative to apprentices. Workers with missing or no qualifications fall by 14.6 log points relative to apprentices, and most of this decline comes from lower mean establishment effects. University workers rise by 5.7 log points, with 3.9 log points coming from mean establishment effects.

Interpretation. The education wage gap widens partly because different education groups increasingly work at different kinds of establishments. This is an important refinement to standard skill-price stories. The table suggests that higher education groups do not simply receive higher portable skill returns; they also gain access to workplaces with higher wage premiums. Conversely, low-education workers are increasingly concentrated in low-premium establishments.

8.14 Table VI: Contribution of Person and Establishment Effects to Wage Variation across Occupations and Industries

Read. Table VI decomposes between-occupation and between-industry wage inequality. For occupations, the change in wage variance is explained by mean person effects, mean establishment effects, and their covariance. Establishment effects and covariance together account for about 70% of the between-occupation increase. For industries, person effects explain 44%, establishment effects 19%, and covariance 42%.

Interpretation. Occupation and industry wage gaps are often interpreted as technology, task, or sector effects. This table shows that workplace allocation is deeply involved. Occupations and

TABLE V
DECOMPOSITION OF CHANGES IN RELATIVE WAGES BY EDUCATION LEVEL, 1985–1991
VERSUS 2002–2009

	(1) Change in mean log wage relative to apprentices	(2) Change in mean person effect	(3) Change in mean establishment effect	(4) Remainder
Highest education qualification				
1. Missing/none	–14.6	1.8	–12.2	–4.2
2. Lower secondary school or less (no vocational training)	–10.5	–0.1	–6.3	–4.1
4. Abitur with or without vocational training*	10.1	0.0	2.6	7.5
5. University or more	5.7	1.5	3.9	0.3

Notes. Wage changes are measured between intervals 1 (1985–1991) and 4 (2002–2009). Remainder (column (4)) represents changing relative contribution of Xb component.
*Abitur refers to Allgemeine Hochschulreife, a certificate of completion of advanced level high school.

Table V: Decomposition of Changes in Relative Wages by Education Level, 1985–1991 versus 2002–2009

TABLE VI
CONTRIBUTION OF PERSON AND ESTABLISHMENT EFFECTS TO WAGE VARIATION ACROSS OCCUPATIONS AND INDUSTRIES

	(1) Interval 1 1985–1991	(2) Interval 2 1990–1996	(3) Interval 3 1996–2002	(4) Interval 4 2002–2009	Change in variance (Int. 1 to Int. 4)*	(6) Share
Panel A: Between occupations (342 three-digit occupations)						
Std. dev. of mean log wages	0.233	0.243	0.263	0.289	0.029	100
Std. dev. of mean person effects	0.186	0.203	0.198	0.207	0.008	28
Std. dev. of mean establ. effects	0.101	0.104	0.124	0.135	0.008	28
Correlation of mean person effects and establ. effects	0.110	0.171	0.238	0.291	0.012	42
Panel B: Between industries (96 two-digit industries)						
Std. dev. of mean log wages	0.173	0.184	0.203	0.224	0.020	100
Std. dev. of mean person effects	0.103	0.114	0.128	0.140	0.009	44
Std. dev. of mean establ. effects	0.104	0.110	0.108	0.121	0.004	19
Correlation of mean person effects and establ. effects	0.242	0.301	0.422	0.403	0.008	42

Notes. Decompositions based on estimated AKM models summarized in Table III. Occupation is based on main job in each year; establishments are assigned one industry per interval, using consistently-coded two-digit industry.
*Entry in column (5) represents change in variance or covariance component. Entry in column (6) is the share of the total change in variance explained. Shares do not add to 100% because Xb component and its covariances are omitted.

Table VI: Contribution of Person and Establishment Effects to Wage Variation across Occupations and Industries

8.15 Figure IX: Standard Deviation of Establishment Effects and Fraction Covered by Collective Agreements, by Birth Year of Establishment

Read. Figure IX plots the standard deviation of establishment effects by establishment birth year and overlays collective bargaining coverage. Newer establishments, especially those entering after the mid-1990s, show greater dispersion in establishment effects. Collective bargaining coverage declines for newer cohorts.

Interpretation. This figure links the wage-premium decomposition to institutional change. The rise in establishment heterogeneity is not mainly driven by old establishments becoming much more heterogeneous. Instead, new establishments enter with more dispersed wage premiums. The decline in collective bargaining coverage among these cohorts is a plausible proximate source of the increased dispersion, because collective bargaining tends to compress establishment wage policies.

8.16 Figure X: Distribution of Establishment Effects by Collective Bargaining Status, Based on Establishment Effects for 1996–2002 and Bargaining Status in 2000 Wave of LIAB

Read. Figure X shows the distribution of establishment-effect deciles by collective bargaining status. Establishments without collective bargaining are overrepresented in the lowest deciles. Establishments with sectoral or firm-specific contracts are more likely to be found in higher establishment-effect deciles.

Interpretation. This figure suggests how declining collective bargaining can increase wage inequality. If uncovered establishments pay lower premiums and have a more dispersed distribution, then growth in the uncovered sector will mechanically widen establishment pay gaps. The result is descriptive rather than causal, because firms choose whether to be covered, but it provides a strong

institutional clue for why establishment heterogeneity rose in West Germany.

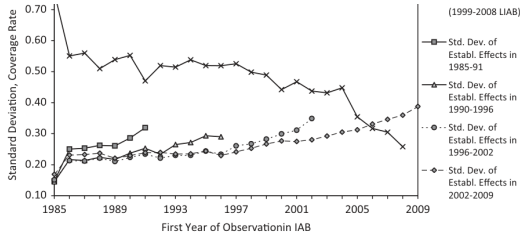


FIGURE IX
Standard Deviation of Establishment Effects and Fraction Covered by Collective Agreements, by Birth Year of Establishment

Figure shows standard deviation of estimated establishment effects in a given observation interval (1985–1991, 1990–1996, 1996–2002, or 2002–2009) for establishments that are present in that interval and first appeared in the IEB data in the “birth year” indicated on the horizontal axis. Figure also shows fraction of establishments in a given birth year surveyed in the 1999–2008 LIAB that are covered by collective agreements.

Figure IX: Standard Deviation of Establishment Effects and Fraction Covered by Collective Agreements, by Birth Year of Establishment

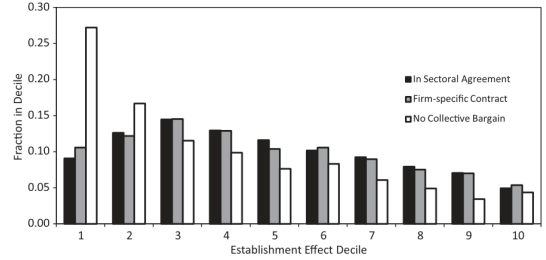


FIGURE X
Distribution of Establishment Effects by Collective Bargaining Status, Based on Establishment Effects for 1996–2002 and Bargaining Status in 2000 Wave of LIAB

Figure shows distribution of collective bargaining coverage status (no collective bargain, covered by firm-specific agreement, or covered by sectoral agreement) for 7,080 establishments in 2000 wave of LIAB that can be linked to IEB data. Establishments are classified into deciles of their estimated establishment effects from AKM model fit to 1996–2002 data.

Figure X: Distribution of Establishment Effects by Collective Bargaining Status, Based on Establishment Effects for 1996–2002 and Bargaining Status in 2000 Wave of LIAB

9 Interpretation of the paper’s mechanism

The paper’s decomposition can be read as a labor-market structure story. There are three channels:

1. **Portable worker inequality.** Workers differ more in the wage component they carry across jobs. This captures conventional skill or worker-level inequality.
2. **Workplace wage-premium inequality.** Establishments differ more in the wage premiums they pay to any worker. This could reflect differences in productivity, rents, management practices, bargaining regimes, or pay policies.
3. **Assortative matching.** High person-effect workers increasingly work at high establishment-effect firms. This makes overall inequality larger than it would be if workers were randomly assigned to firms.

The third channel is particularly important conceptually. Even if the distributions of person effects and establishment effects were fixed, the wage distribution would become more unequal if high- α workers were increasingly matched to high- ψ establishments. The paper shows that the real world combines all three forces.

10 Limitations and cautions

The AKM model is powerful, but its interpretation still depends on assumptions. Establishment effects are identified from movers, so if mobility is driven by unobserved match quality or transitory shocks in ways not captured by the diagnostics, estimates could be biased. The authors provide several checks, but no fixed-effect decomposition can fully eliminate all concerns about endogenous mobility.

The collective bargaining evidence is also descriptive. Establishments choose whether to enter collective agreements, and that choice may be correlated with productivity, management, worker composition, or local labor market conditions. The paper therefore presents declining collective bargaining coverage as a proximate institutional correlate, not as a fully identified causal mechanism.

Finally, the paper studies West Germany, a setting with distinctive labor-market institutions and reunification-era reforms. The mechanisms are likely relevant elsewhere, but the relative importance of establishment effects and sorting may differ across countries depending on wage bargaining institutions, firm productivity dispersion, and labor mobility.

11 Bottom line

The paper's bottom line is that rising wage inequality in West Germany is a joint product of workers, firms, and matching. Worker inequality rises, workplace wage-premium inequality rises, and high-wage workers increasingly sort into high-premium workplaces. This makes establishment heterogeneity and assortative matching central to understanding modern wage inequality.